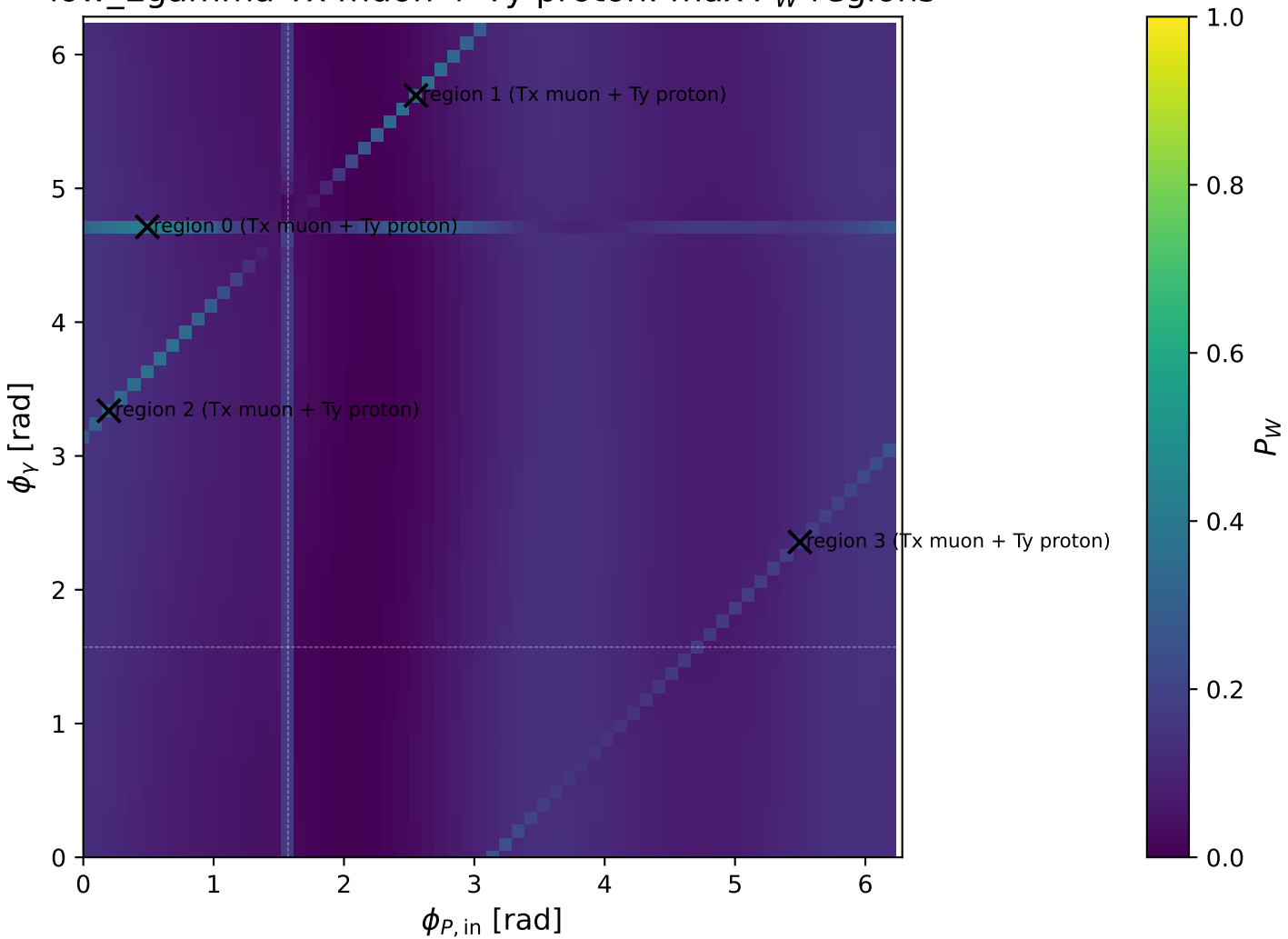
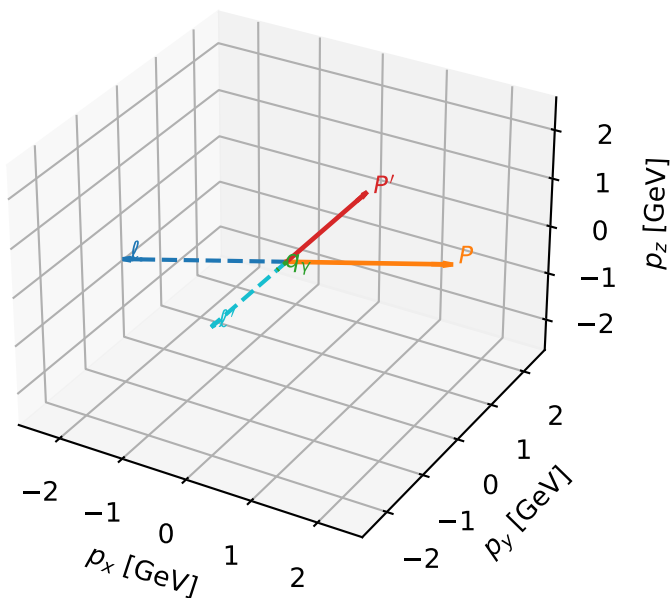


low_Egamma Tx muon + Ty proton: max P_W regions



Tx muon + Ty proton characteristic max P_W configuration

3D momenta

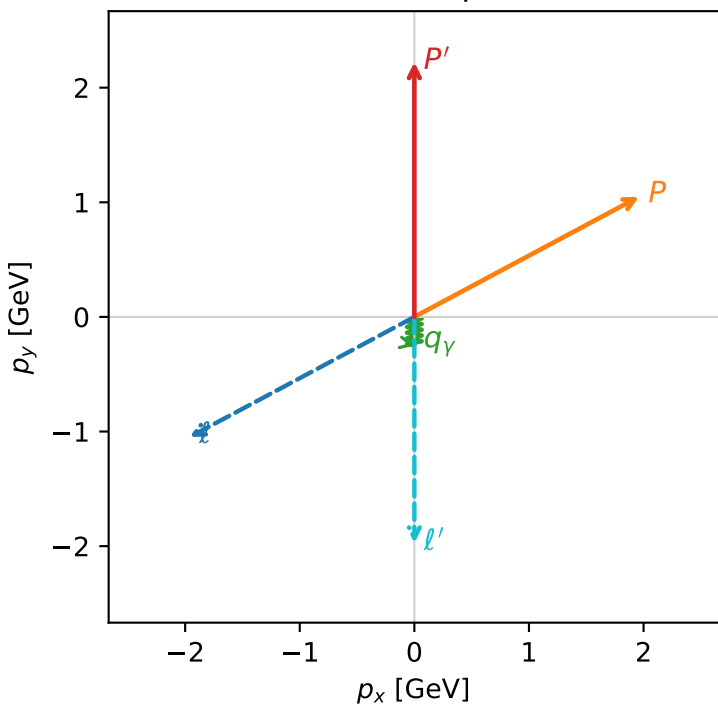


w_purity_low_egamma_txtty_region_0 P_W=0.412937
 region: low_Egamma_TxTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

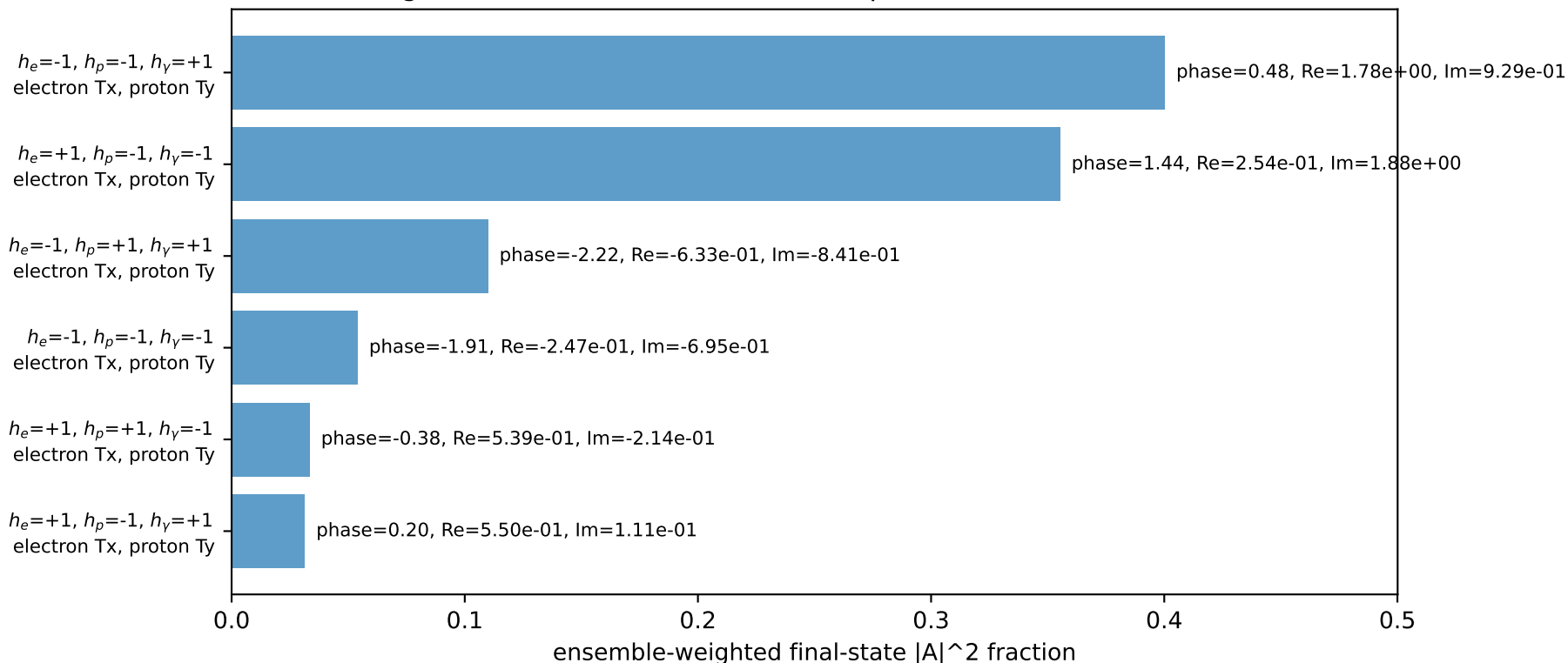
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.22295$
 $\theta_{in}=1.57079$, $\phi_l=3.63247$, $\phi_P=0.490874$
 $E_Y=0.25$, $\phi_Y=4.71239$
 $Q^2=4.63715$, $x_B=0.666737$, $t=-5.22456$, $W^2=3.19769$, $y=0.337215$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -1.9606$ $p_y= -1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= 1.0480$ $p_z= 0.0000$
 ℓ' $E= 1.9758$ $p_x= 0.0000$ $p_y= -1.9729$ $p_z= 0.0000$
 P' $E= 2.4127$ $p_x= 0.0000$ $p_y= 2.2229$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

Transverse plane

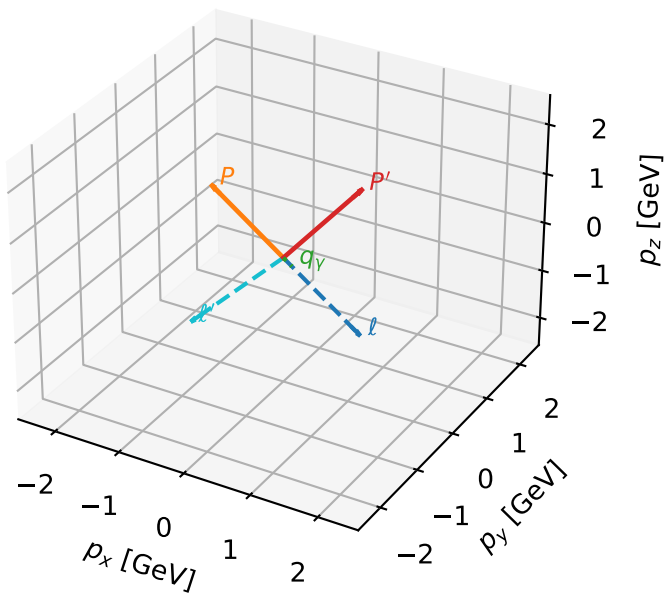


Leading initial-ensemble/final-state components (N=6, retained=98.3%)



Tx muon + Ty proton characteristic max P_W configuration

3D momenta

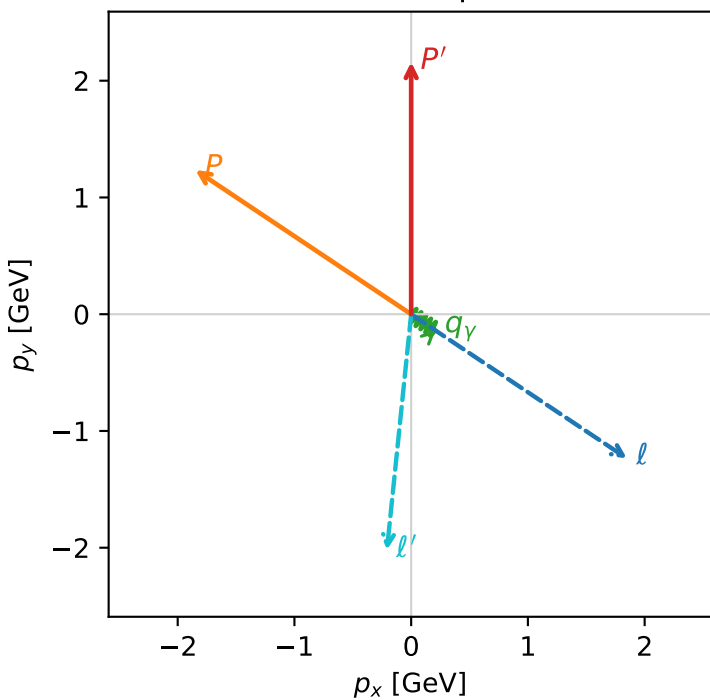


w_purity_low_egamma_txtty_region_1 P_W=0.355039
 region: low_Egamma_TxTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

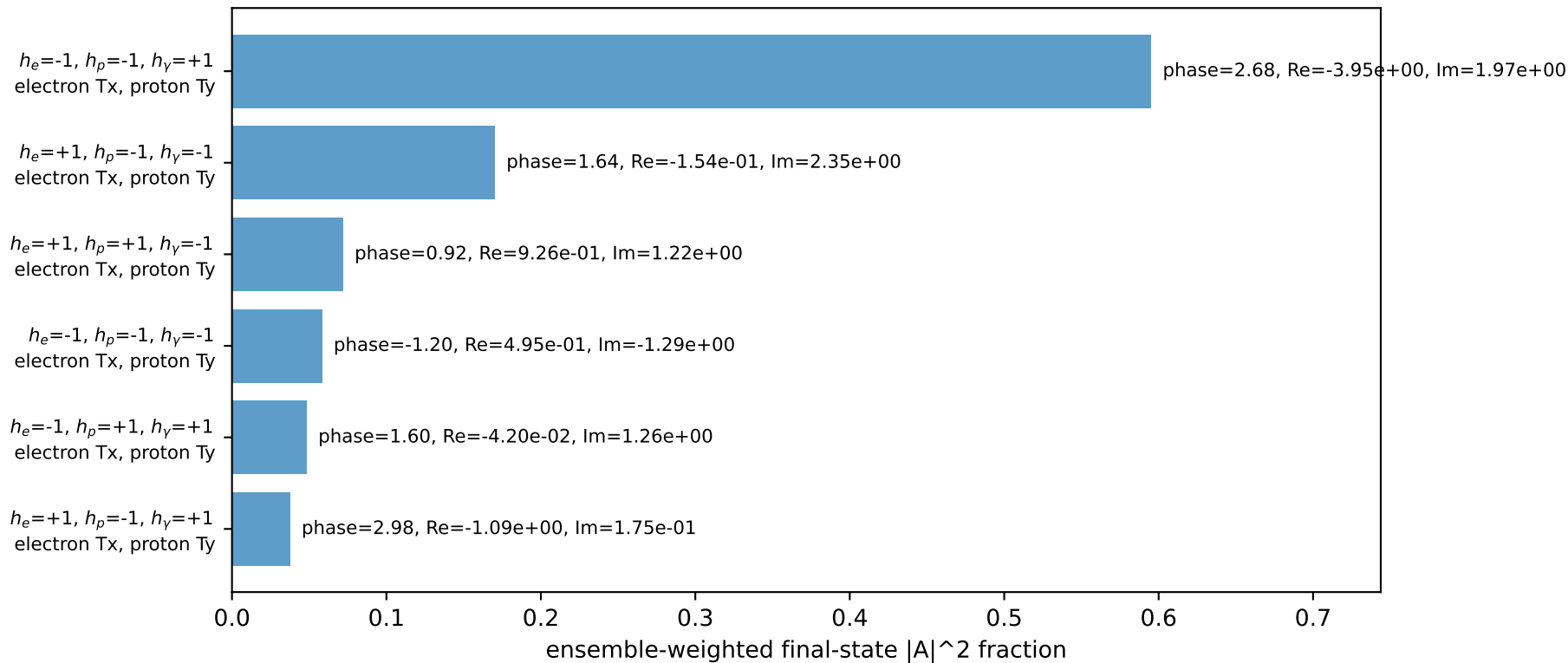
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.15954$
 $\theta_{in}=1.57079$, $\phi_l=5.69414$, $\phi_p=2.55254$
 $E_\gamma=0.25$, $\phi_\gamma=5.69414$
 $Q^2=4.80884$, $x_B=0.730166$, $t=-4.26796$, $W^2=2.65696$, $y=0.319322$
 $\theta(l', \gamma)=62.1234$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.8484$ $p_y= -1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.8484$ $p_y= 1.2351$ $p_z= 0.0000$
 l' $E= 2.0341$ $p_x= -0.2079$ $p_y= -2.0207$ $p_z= 0.0000$
 P' $E= 2.3545$ $p_x= 0.0000$ $p_y= 2.1595$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2079$ $p_y= -0.1389$ $p_z= 0.0000$

Transverse plane

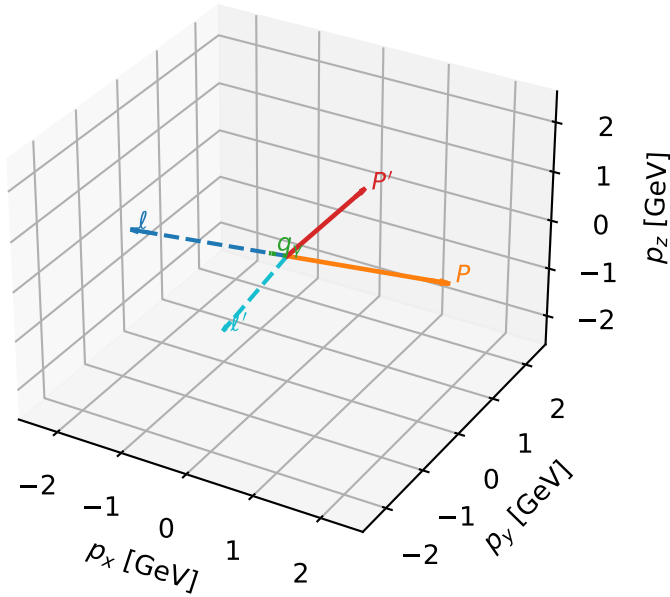


Leading initial-ensemble/final-state components (N=6, retained=98.1%)



Tx muon + Ty proton characteristic max P_W configuration

3D momenta

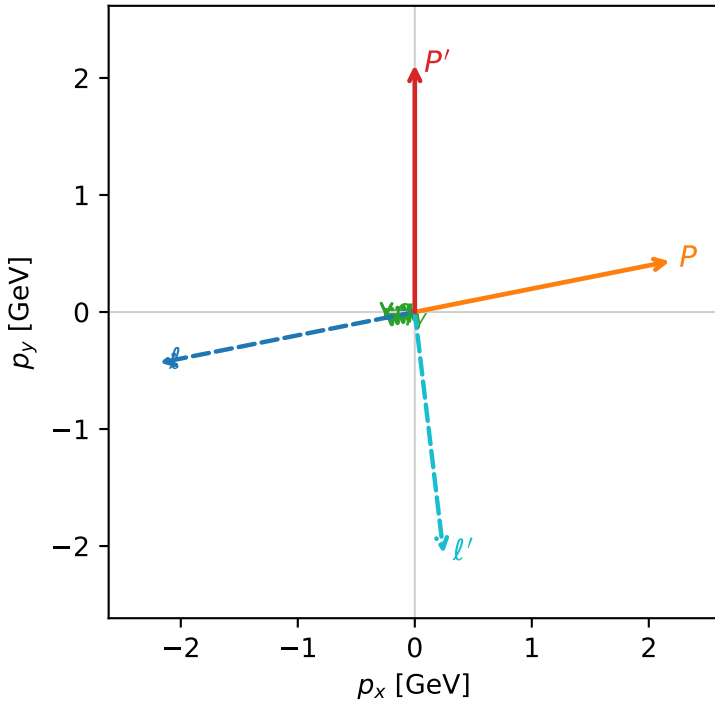


w_purity_low_egamma_txtty_region_2 P_W=0.312443
 region: low_Egamma_TxTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

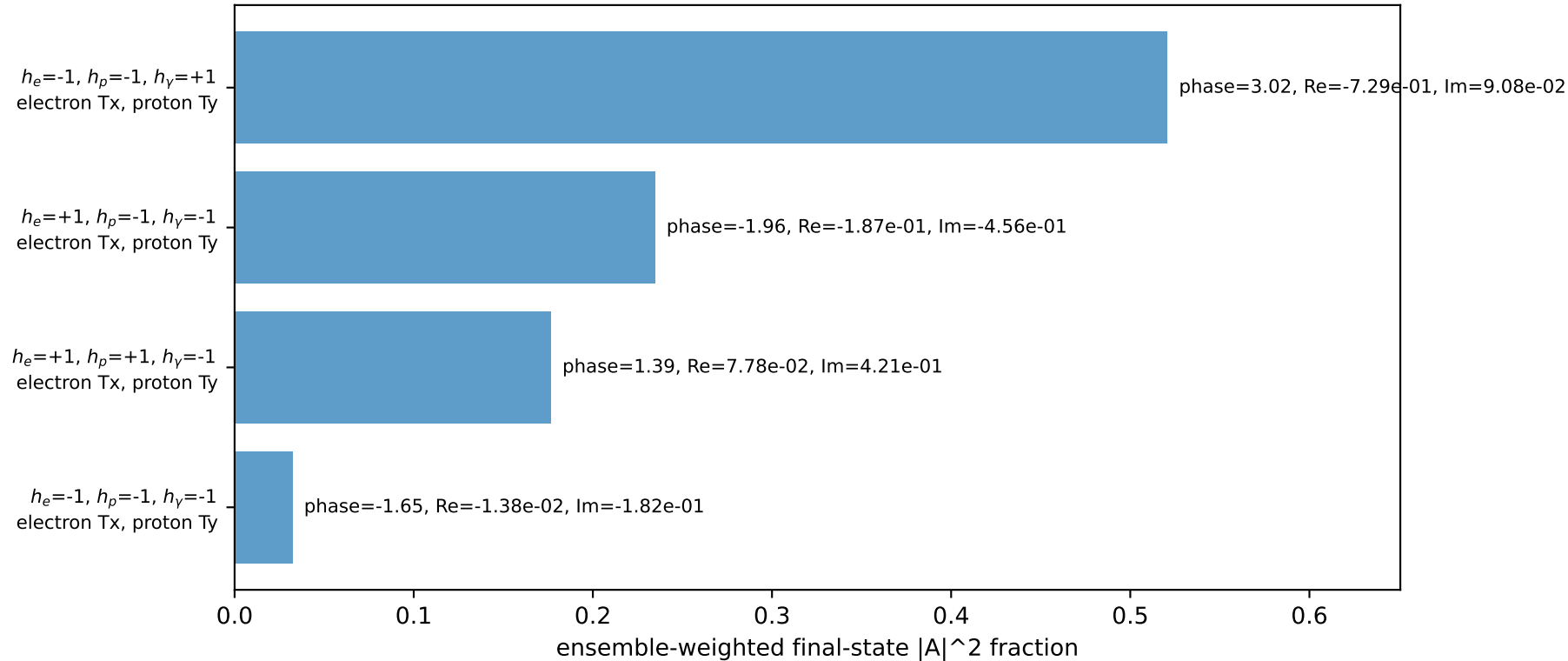
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.11051$
 $\theta_{in}=1.57079$, $\phi_l=3.33794$, $\phi_P=0.19635$
 $E_\gamma=0.25$, $\phi_\gamma=3.33794$
 $Q^2=8.51246$, $x_B=0.862186$, $t=-7.55511$, $W^2=2.2405$, $y=0.4787$
 $\theta(l', \gamma)=85.5322$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -2.1804$ $p_y= -0.4337$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 2.1804$ $p_y= 0.4337$ $p_z= 0.0000$
 l' $E= 2.0790$ $p_x= 0.2452$ $p_y= -2.0617$ $p_z= 0.0000$
 P' $E= 2.3096$ $p_x= 0.0000$ $p_y= 2.1105$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2452$ $p_y= -0.0488$ $p_z= 0.0000$

Transverse plane

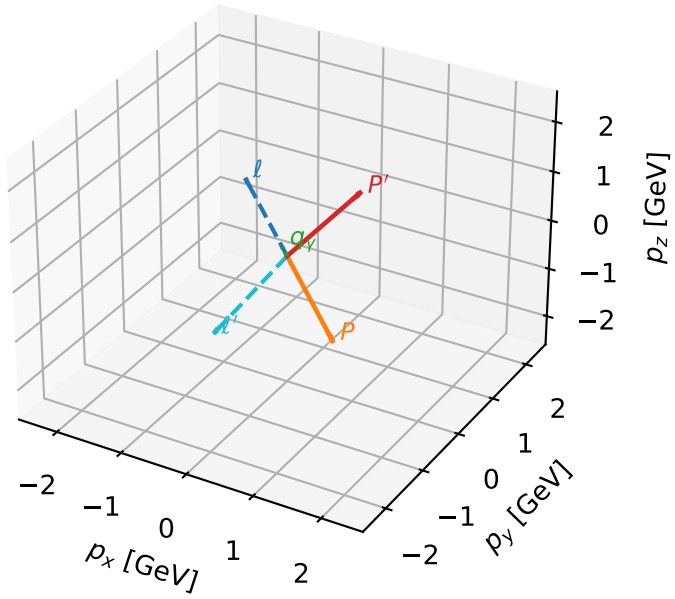


Leading initial-ensemble/final-state components (N=4, retained=96.4%)



Tx muon + Ty proton characteristic max P_W configuration

3D momenta

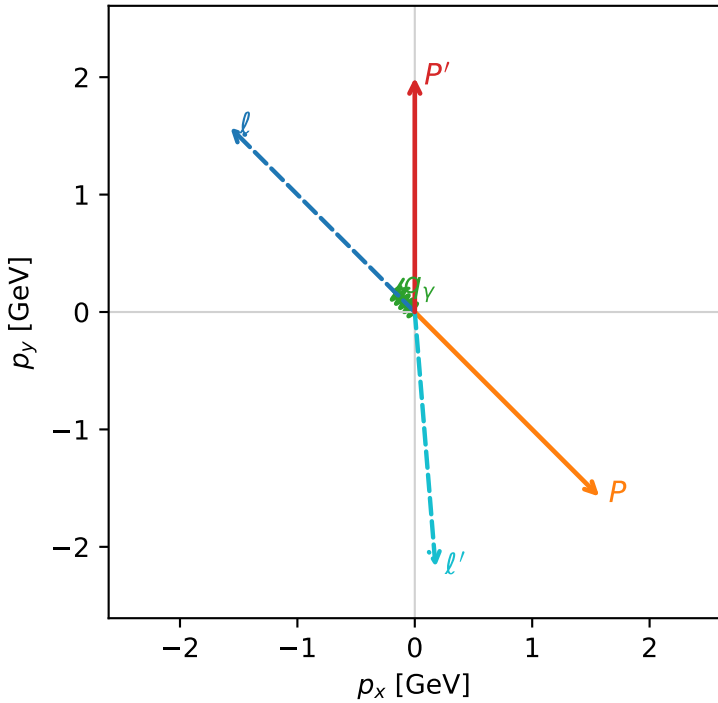


w_purity_low_egamma_txtty_region_3 P_W=0.179824
 region: low_Egamma_TxTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.99631$
 $\theta_{in}=1.57079$, $\phi_l=2.35619$, $\phi_p=5.49779$
 $E_Y=0.25$, $\phi_Y=2.35619$
 $Q^2=17.0818$, $x_B=0.977284$, $t=-15.1608$, $W^2=1.27689$, $y=0.847466$
 $\theta(l', \gamma)=139.651$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.5720$ $p_y= 1.5720$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.5720$ $p_y= -1.5720$ $p_z= 0.0000$
 l' $E= 2.1828$ $p_x= 0.1768$ $p_y= -2.1731$ $p_z= 0.0000$
 P' $E= 2.2057$ $p_x= 0.0000$ $p_y= 1.9963$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.1768$ $p_y= 0.1768$ $p_z= 0.0000$

Transverse plane



$h_e=-1$, $h_p=-1$, $h_Y=+1$
 electron Tx, proton Ty

$h_e=+1$, $h_p=+1$, $h_Y=-1$
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=97.3%)

